

GUI) and stands as an intermediary between a digital video source, like an AVI or MPEG file, and a VFW receiving program, which is typically a media player, video editing software or encoder.

AviSynth communicates with any program that supports AVIs through the Video for Windows system by acting as a fake AVI file. Using AviSynth users can apply a large variety of editing and processing functions, called filters, to a video stream before passing along the results. Filter capabilities include trimming, cropping, deinterlacing, inverse telecine, still image loading and splicing, colour corrections and denoising.

The scripting language is a programming language containing many features familiar to programmers, including variables, distinct data types and complex expressions. The language deals primarily with videos, as a primitive data type. A typical script loads a video as input, applies processing to it, then returns that video in the same way as functions return values. The returned video is the output of the script, which is seen as the video by the program running the script.

It also has several other more standard data types, including int, float, bool and string. These can be used to perform calculations and write text such as subtitles to the video.

Csound

Csound is a computer programming language for sound, also known as sound compiler, audio programming language or, more precisely, C-based audio DSL. It takes two specially formatted text files as input. Orchestra describes the nature of the instruments and score describes notes and other parameters along a timeline. Csound processes instructions in these files and renders an audio file or real-time audio stream as output. It can be used along with an IDE for easier development of the program.

The use of plug-ins allows additional capabilities without modifica-



Fig. 1: Audacity in action

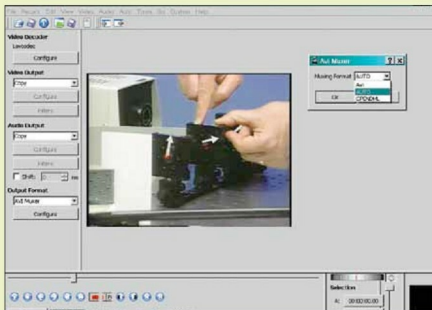


Fig. 2: Video track editing in Avidemux

tions to the Csound code, as there is the possibility to write user-defined op-codes as extensions to the original language. Supported plug-ins include LADSPA, DSSI and VST.

FFmpeg

FFmpeg is a free software project that produces libraries and programs for handling multimedia data. It includes libavcodec (an audio/video codec library used by several other projects), libavformat (a library for dealing with

various media container formats) and ffmpeg command line program for transcoding multimedia files. It is used in many video projects such as VLC media player, Youtube, Mplayer, Blender and Google Chrome.

Jokosher

Jokosher is a free software for non-linear multi-track digital audio editing. It is developed in Python, using GTK+ interface and GStreamer as an audio back-end. It has a very user-